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Studierichting: Informatica
Oefeningenreeks 5I nr. 1.2

$f(x) = 2\sqrt{x}$

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$L = \int_0^1 \sqrt{1 + \left(\left(2\sqrt{x}\right)'\right)^2} dx$

$\int \sqrt{1 + \left(\left(2\sqrt{x}\right)'\right)^2} dx$

$= \int \sqrt{1 + \left(\frac{1}{\sqrt{x}}\right)^2} dx$

$= \int \sqrt{1 + \frac{1}{x}} dx$

$= \int \frac{\sqrt{x+1}}{\sqrt{x}} dx$

Substitutie: $x = t^2 \Rightarrow dx = 2tdt \Leftrightarrow t = \sqrt{x}$

$= \int \frac{\sqrt{t^2+1}}{t} 2tdt$

$= 2 \int \sqrt{t^2+1} dt$

(Hyperbolic) substitutie: $t = \sinh u \Rightarrow dt = \cosh u du$

$$= 2\int \sqrt{\sinh^2 u + 1}\cosh u du$$

$$= 2\int \cosh^2 u du$$

$$= 2\int \frac{\cosh 2u + 1}{2} du$$

$$= \int (\cosh 2u + 1)\, du$$

$$= \frac{\sinh 2u}{2} + u + c$$

$$= \sinh u \cosh u + u + c$$

$$\overset{\cosh u = \sqrt{t^2+1},\, u = \mathrm{Bgsinh}\, t}{=} t\sqrt{t^2+1} + \ln|t + \sqrt{t^2+1}| + c$$

$$= \sqrt{x}\sqrt{x+1} + \ln|\sqrt{x} + \sqrt{x+1}| + c$$

$$\Rightarrow L = \left[\sqrt{x}\sqrt{x+1} + \ln|\sqrt{x} + \sqrt{x+1}|\right]_0^1$$

$$= \sqrt{2} + \ln|1 + \sqrt{2}| - \ln 1$$

$$= \sqrt{2} + \ln\left(1 + \sqrt{2}\right)$$

References